

CONFIDENTIAL · EXECUTIVE BRIEFING · RISK COMMITTEE

Strategic Credit Risk & Portfolio Analytics Report

What 6,985 loan applications, 13,000 payments, and a formal data-validation pass reveal about how this bank actually makes lending decisions.

₹418 Cr	4,068	66	82
Approved Portfolio	Funded Loans	Branches	SQL Queries Validated

BankLoanCreditRisk Portfolio | Customers · Loans · Payments · Branches · Loan Types
Prepared for Executive Leadership | FY 2025-26

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Everything in this report traces back to **82 queries across six SQL scripts: 01_Database_Setup_Validation** through **06_Business_Insights**. Portfolio metrics are scoped to the `dbo.Approved_Loans` view (`Approval_Status = 'Approved'`); funnel metrics use the full `dbo.Loans` table. Each table in this report says which one it's drawing from.

01 Executive Summary

Walk into this portfolio cold and the headline numbers look fine. Look one layer deeper, and three things stop looking fine and start looking structural.

The bank has **4,500 customers** and has taken in **6,985 loan applications**, of which **4,068 (58.2%) were funded**, **2,865 (41.0%) were turned down**, and **52 are still sitting in the queue** (all from 2024, a live backlog, not old business). That funded book, the `Approved_Loans` view this whole report is built on, carries **₹418.37 crore** in face value, **₹295.17 crore** of it still outstanding. It defaults at **5.01%** and reaches NPA at a further **2.46%**, for a combined 7.47%. That's the number that will keep coming up throughout this report as the baseline everything else gets compared against.

4,068 Approved Loans	₹418.37 Cr Total Approved Value	5.01% / 2.46% Default Rate / NPA Rate	₹295.17 Cr Outstanding Balance
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■ The Three Things That Matter Most

Each of these came out of the validation and risk-analysis work in Sections 2 and 5. They're presented here because they change how much confidence the rest of this report can carry, and every downstream pricing, provisioning, or collections decision along with it.

74% combined failure rate inside loans tagged "Critical"	Risk_Category doesn't predict risk — it announces it after the fact Split the approved book by <code>Risk_Category</code> and something odd shows up: Low, Medium, and High together produce a flat 0.00% Default rate and 0.00% NPA rate across 3,658 loans. Not low — zero. Every single failure in the portfolio sits inside the 410 loans tagged "Critical," where 49.76% have already defaulted and another 24.39% have gone NPA. A category meant to separate future risk from future safety is really just relabelling loans that have already failed. The frustrating part: a working alternative is already sitting in the data. Bin <code>Credit_Score</code> into the five standard bureau bands and default-plus-NPA climbs cleanly from 0.78% in the top band to 31.6% in the bottom. That's a 40x spread, using only information the bank has at the moment someone applies.
32x higher combined default rate, Sub-Prime vs. Premium borrowers	The bank is gatekeeping by segment, not by risk Sub-Prime applicants get approved 27.8% of the time, against 93.3% for Premium and 86.5% for Mass Affluent. On its face, that looks like caution. But the Sub-Prime loans that do get through still default at 19.2% combined: eleven times the Mass Affluent rate, thirty-two times the Premium rate. If the 27.8% who clear the bar were being underwritten more carefully than the 72.2% who don't, their outcomes should look better than the segment average. They don't. The gate is filtering volume, not risk, and it's not clear the bank is getting the credit-quality benefit that a 72% rejection rate should be buying it.

100%

of approved loans fail the
Amount_Repaid reconciliation
check

The two ledgers that should agree on repayment, don't

Q13 checks something simple: does each loan's Amount_Repaid field match what the Payments table shows was actually paid, give or take a rupee. It fails for all 4,068 approved loans. Not most — all of them.

On average, Amount_Repaid claims 5.25x more than the Payments table can account for, and the two figures only correlate at 0.75: too loose for a rounding issue, too consistent to be random. The more likely explanation is that Amount_Repaid is coming from a different system, possibly a scheduled-repayment projection rather than actual cash received. Until that's resolved, every outstanding-balance and collections figure downstream, including the ₹295.17 crore reported above as outstanding, should be treated as provisional.

■ What We'd Do First

Two things are foundational enough that everything else should wait on them: **rebuild Risk_Category into a genuine pre-approval score**, starting from the credit-band gradient already visible in the data, and **treat the Payments/Loans mismatch as a data-integrity incident**, not a footnote, given it touches literally every approved loan. Section 9 lays out five recommendations in total; these two come first because every pricing, provisioning, and collections number in this report ultimately rests on them.

02 Data Quality & Validation

A risk report is only as good as the data underneath it. Before trusting a single default rate in this document, the updated SQL suite runs the whole database through a formal check: Q9 through Q13 in 01_Database_Setup_Validation.sql. Most of it comes back clean. One part of it doesn't, and that part changes how the rest of the numbers in this report should be read.

■ 2.1 Does everything point where it should? (Q9–Q10)

Referential integrity first: does every loan point to a customer, branch, and loan type that actually exists, and does every payment point to a real loan? Customers, Loans, and Branches all check out clean.

TABLE 2.1 — REFERENTIAL INTEGRITY & PRIMARY KEYS

Check	Result
Loans referencing a missing Customer_ID	0
Loans referencing a missing Branch_ID	0
Loans referencing a missing Loan_Type_ID	0
Payments referencing a missing Loan_ID	2 — both point to Loan_ID LN0005644, which does not exist
Duplicate Customer_ID	0
Duplicate Loan_ID	0

The one thing that doesn't check out is small in volume but worth flagging on principle: two Payment records reference a Loan_ID that was never issued. It's a minor finding on its own, but it turns out to be a preview of a much bigger reconciliation problem in 2.4.

■ 2.2 Are the core numbers even usable? (Q11)

Zero loans have a null, zero, or negative Loan_Amount or Interest_Rate. The two fields nearly every calculation in this report depends on are complete and well-formed. That's the good news this section needed.

■ 2.3 Why this report can't just query dbo.Loans directly (Q12)

Cross-tabulating Approval_Status against Loan_Status shows exactly why every portfolio metric in this report is scoped to approved loans specifically, rather than the raw Loans table:

TABLE 2.2 — LOAN_STATUS BY APPROVAL_STATUS

Approval_Status	Loan_Status	Count
Approved	Active	2,866
Approved	Closed	898
Approved	Default	204
Approved	NPA	100
Pending	Rejected	52
Rejected	Rejected	2,865

Notice the Pending row: 52 applications that haven't been decided yet are stamped with the same 'Rejected' placeholder as the loans that were genuinely turned down. They're not actually rejected; the field just has nowhere else to point until a decision is made. Loan_Status, in other words, only carries real meaning once a loan is Approved. Group by it without filtering to Approval_Status = 'Approved' first, and 2,917 applications that were never funded quietly blend into what should be a funded-portfolio metric. That's exactly the mistake dbo.Approved_Loans (defined in this script, used from 02 onward) exists to prevent. It's why every figure in this report carries that same scope.

■ 2.4 Do the repayment numbers actually add up? (Q13)

This is the one that matters most. Q13 takes every approved loan's Amount_Repaid field and checks it against what the Payments table actually shows was collected for that loan, allowing a one-rupee tolerance for rounding.

100%

of approved loans (4,068 of 4,068)
fail this check

Not a partial gap — a systemic one

The median gap between what Amount_Repaid claims and what Payments can evidence is ₹127,678 per loan. Averaged across the book, Amount_Repaid runs to ₹290,416 per loan against ₹55,286 actually visible in Payments, 5.25 times higher, and the two figures correlate at only 0.75.

A partial data extract would explain a gap that's consistent or concentrated in recent loans. It would not explain a 100% failure rate with only moderate correlation between the two fields. The more plausible read is that Amount_Repaid is being populated from somewhere else entirely: a scheduled or expected-repayment figure, not actual cash collected.

One more layer to this: 499 approved loans (12.3%) have no Payments records at all, and that includes 130 of the 304 Default-and-NPA loans. That's 42.8% of the entire failed book with no transactional trail explaining how it got there.

Of the ten loans with the largest repayment gaps, eight are marked 'Closed,' recorded as fully settled despite payment records covering as little as 0–21% of the reported amount.

Practically, this means **Amount_Repaid, Outstanding_Balance, and any collections-efficiency figure derived from them should be treated as provisional** until Finance and Data Engineering can trace the actual source system. Recommendation 2 in Section 9 spells out exactly what that investigation should look like.

03 Power BI Dashboard — Structure & Coverage

The “Bank Loan Credit Risk Analytics” Power BI file organizes the same portfolio into four linked pages that a branch manager or committee member would actually click through day to day. The four captures below are pulled directly from the live report, not a recreation, so what's on screen is whatever the dashboard's own DAX measures produce.

That turns out to matter. Cross-checking every figure on these pages against the source tables, everything ties out **except one thing worth flagging**: several KPI cards on the same strip use different denominators without saying so. “Approved Loans: 4,068” sits right next to “NPA Ratio: 1.43%,” but 1.43% is $100 \div 6,985$ (all applications), not $100 \div 4,068$ (the approved book, which would read 2.46%, the figure used throughout this report). Same story for Default Rate (2.92% here vs. 5.01% on an approved-only basis) and Average Interest Rate (10.26% here vs. 10.03% approved-only). Neither number is wrong (they're just answering different questions), but shown side by side with no label distinguishing them, it's an easy misread for anyone skimming the dashboard the way it's meant to be used. It's a smaller-scale version of the exact Loan_Status issue in Section 2.3, playing out in the tool people actually look at.

■ 3.1 Executive Overview



FIGURE 3.1 Eight KPI cards (four scoped to all 6,985 applications, four to the approved book, see note above), Loan Portfolio by Risk Category, Approval vs. Rejection Trend, Loan Status Breakdown, Default Rate by Branch, Monthly Disbursement Trend, Top 10 Customers by Loan Value, and a four-tile Risk Summary. The Risk Summary tiles (769 / 1,314 / 1,575 / 410) match this report's Section 5.4 figures exactly.

■ 3.2 Customer Analysis



FIGURE 3.2 Customer Segment, Occupation, and Education distributions; Avg Annual Income by Segment; Loan Amount by Customer Segment; Customer Distribution by region; Top 10 Customers by Credit Profile. The segment bars (2,752 / 1,077 / 561 / 110) match Section 4.3 exactly, as does the credit-profile table.

3.3 Risk Analysis



FIGURE 3.3 Risk Category Distribution by loan value, Loan Status Trend, Default Analysis, NPA Analysis, Outstanding Balance by loan type, Amount Repaid, High-Risk Customers, and Risk by Loan Type. Total Outstanding Balance reads ₹295.17 Cr, an exact match to Section 4.2. Total NPA Amount (₹8.09 Cr) is the outstanding balance specifically on NPA loans, not their original loan amount (₹8.84 Cr, Section 4.2); a different and equally valid cut, just worth not conflating the two.

3.4 Branch Analysis



FIGURE 3.4 Branch/Region map, Region Performance, Branch-wise Loan Amount, Regional Loan Distribution, Branch NPA Rate, Loan Count by Branch, Approval vs. Default Comparison, and a Branch Performance table. Best Region reads North, matching Section 5.3. Top Performing Branch reads Fraser Road: highest approval rate in the network (69.0%) despite carrying an internal 'Low' Performance_Tier label, the same disconnect documented in Section 5.8.

04 Portfolio Composition & Structure

Before asking where risk concentrates, it's worth understanding what the book is actually made of: who the bank lends to, how big the branch network is, and how quickly applications move through it.

■ 4.1 The application funnel (dbo.Loans)

Of every 100 applications that come in, 58 get funded, 41 get turned away, and roughly one is still waiting on a decision:

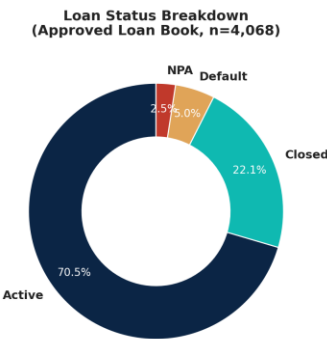
TABLE 4.1 — APPLICATION FUNNEL

Outcome	Count	% of Total
Approved	4,068	58.2%
Rejected	2,865	41.0%
Pending	52	0.7%

■ 4.2 What the funded book looks like today (dbo.Approved_Loans)

TABLE 4.2 — APPROVED BOOK BY STATUS

Status	Count	% of Approved	Total Loan Amount
Active	2,866	70.5%	₹352.27 Cr
Closed	898	22.1%	₹39.00 Cr
Default	204	5.0%	₹18.26 Cr
NPA	100	2.5%	₹8.84 Cr



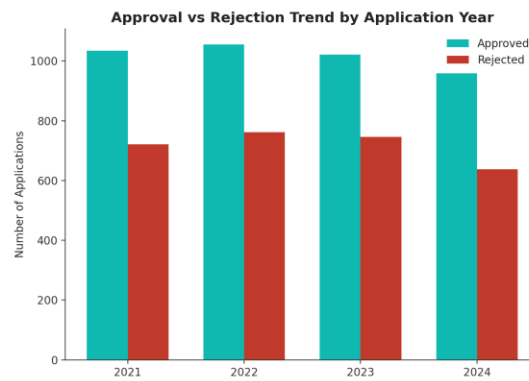


FIGURE 4.2 Left: approved-book status mix. Right: approval vs. rejection volume by application year. Approval volume has held broadly flat since 2021, and the 2024 cohort still carries that 52-application backlog.

■ 4.3 Who the customers are

The 4,500-customer base skews younger than you might expect for a lending book (an average age of 38.1, ranging 19 to 78) and splits close to evenly by gender, 55% male to 45% female. Where it doesn't split evenly is income segment:

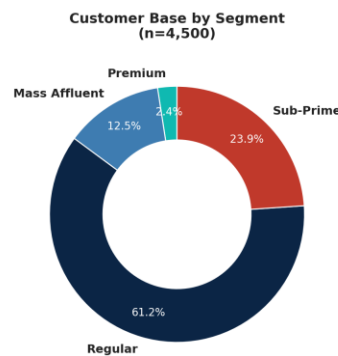


FIGURE 4.3 Regular and Sub-Prime together make up 85.1% of the customer base; Premium is a thin 2.4% slice.

- Regular — 2,752 customers (61.2%), averaging ₹7.75 lakh a year
- Sub-Prime — 1,077 customers (23.9%), averaging ₹5.69 lakh a year
- Mass Affluent — 561 customers (12.5%), averaging ₹14.22 lakh a year
- Premium — 110 customers (2.4%), averaging ₹26.65 lakh a year

There's a real gap between Regular and Mass Affluent: an 83% jump in income with nothing in between, and no product designed to bridge it. Right now, a Regular customer's only path to Mass Affluent is their own income growing on its own; the bank has no upgrade product engineered to move them there.

■ 4.4 The branch network

66 branches, rated Medium (28), Low (21), or High (17) on an internal performance tier (more on whether that tier means what it says in Section 5.8). The single biggest concentration of risk sits with MG Road Branch, carrying ₹11.35 crore in outstanding balance on its own, followed by Bandra, T Nagar, Mall Road, and Park Street. Between them, those five branches are large enough that a shock at any one would move the bank's portfolio-level numbers, not just that branch.

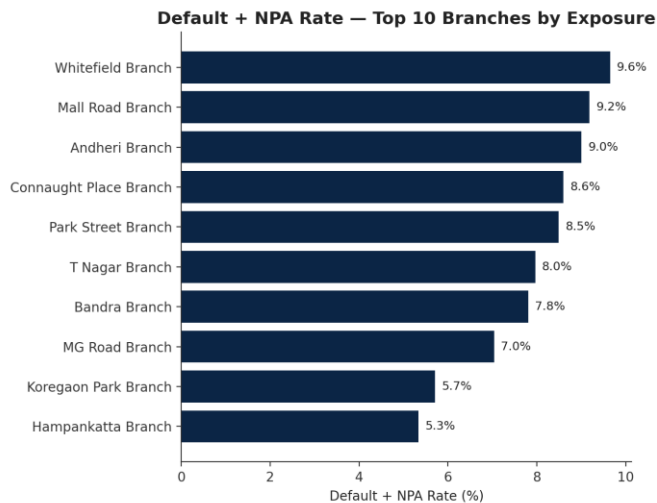


FIGURE 4.4 Default + NPA rate for the ten branches carrying the largest outstanding-balance exposure.

■ 4.5 How long approval actually takes (Q29)

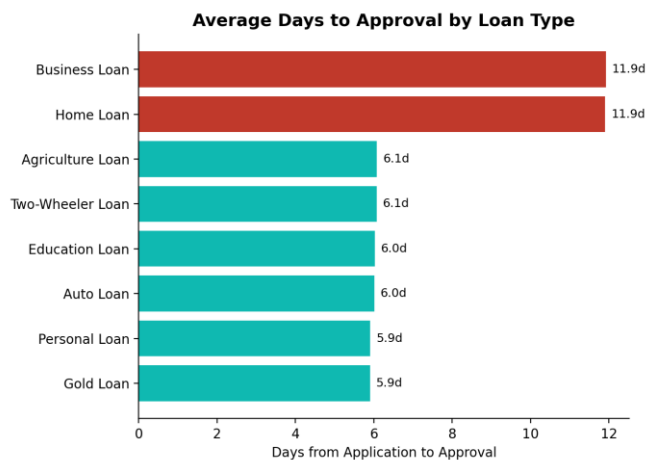


FIGURE 4.5 Average days from application to approval, by loan type.

Approval speed splits into exactly two tiers, with nothing in between. **Business and Home loans take about 11.9 days** to clear; every other product (Personal, Auto, Education, Two-Wheeler, Gold, Agriculture) clears in **5.9 to 6.1 days**, almost exactly half. That's too clean a split to be coincidence; it reads like a deliberate two-track process, larger or secured loans routed to manual underwriting. That's a reasonable design choice, though worth confirming it's actually intentional rather than just queue congestion on the bank's two highest-value products.

05 Risk & Default Deep-Dive

Every table below runs on `dbo.Approved_Loans`, with `Default` and `NPA` reported separately (matching Q69/70 and Q74/75) because the two behave differently depending on how you cut the book. Combined, the book-wide rate is 7.47% (5.01% `Default` plus 2.46% `NPA`, 304 of 4,068 loans). What follows is that same 7.47% broken apart eight different ways to find out where it actually lives.

■ 5.1 By customer segment — still the sharpest line in the data

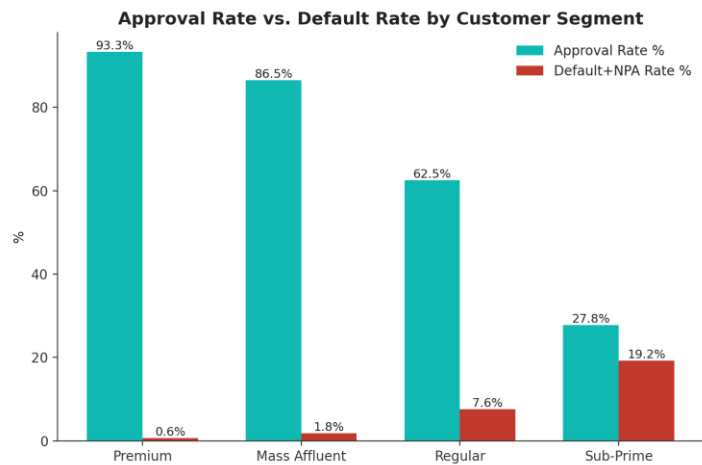


FIGURE 5.1 Approval rate vs. combined default rate, by customer segment.

TABLE 5.1 — DEFAULT & NPA RATE BY CUSTOMER SEGMENT

Segment	Approved Loans	Default Rate	NPA Rate	Approval Rate
Sub-Prime	458	12.88%	6.33%	27.8%
Regular	2,649	5.10%	2.49%	62.5%
Mass Affluent	794	1.13%	0.63%	86.5%
Premium	167	0.60%	0.00%	93.3%

One detail worth sitting with: Premium's entire 0.60% failure rate is `Default`. Not a single loan in that segment has ever reached `NPA`. Combined, Sub-Prime's 19.2% is eleven times the Mass Affluent rate and thirty-two times the Premium rate: by a wide margin, the single biggest risk lever in this portfolio.

■ 5.2 By loan type

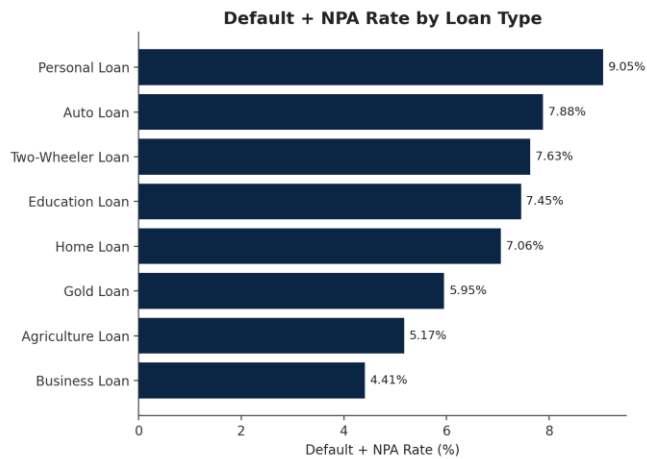


FIGURE 5.2 Combined default + NPA rate by loan type.

TABLE 5.2 — DEFAULT & NPA RATE BY LOAN TYPE

Loan Type	Default Rate	NPA Rate
Personal Loan	6.15%	2.90%
Auto Loan	5.14%	2.73%
Two-Wheeler Loan	5.34%	2.29%
Education Loan	4.81%	2.64%
Home Loan	4.93%	2.13%
Gold Loan	2.38%	3.57%
Agriculture Loan	2.87%	2.30%
Business Loan	3.58%	0.83%

Gold Loan is the one that runs backwards from the rest: its Default rate (2.38%) is among the lowest of any product, but its NPA rate (3.57%) is the highest. Once a Gold Loan goes bad, it seems to move to NPA faster than it lingers in Default, plausible given that gold as collateral makes early write-off or recovery a cleaner call than it is for unsecured products.

■ 5.3 By region

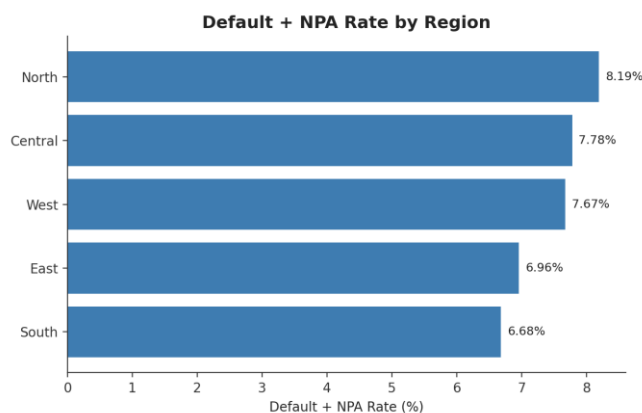


FIGURE 5.3 Combined default + NPA rate by region.

TABLE 5.3 — DEFAULT & NPA RATE BY REGION

Region	Default Rate	NPA Rate
North	5.29%	2.90%
Central	5.93%	1.85%
West	5.15%	2.52%
East	4.97%	1.99%
South	4.39%	2.29%

Regions cluster far more tightly than segments do. Central's combined 7.78% sits well within a point of most of the others. The pattern here comes down to who the borrower is, not where the branch sits.

■ 5.4 By risk category — the circularity problem, in numbers

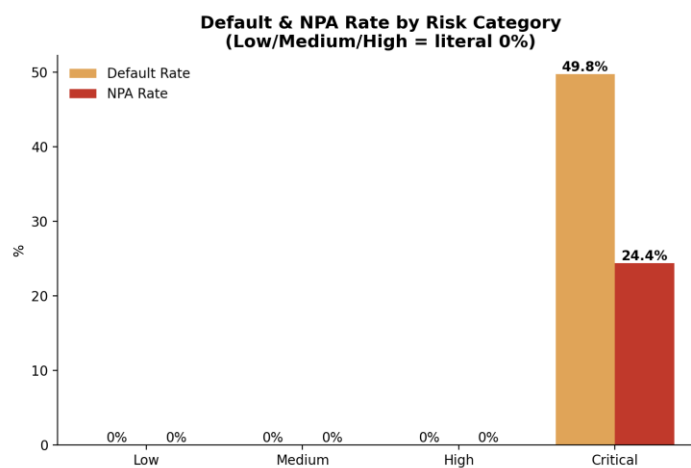


FIGURE 5.4 Default & NPA rate by Risk_Category. Low, Medium, and High are a flat 0%.

Low, Medium, and High together show a literal 0.00% Default rate and 0.00% NPA rate across all 3,658 loans in those three buckets. **Critical** alone carries **49.76% Default** and **24.39% NPA**: 74.15% combined, on just 410 loans. That's not risk-weighting; a category is either failed or it isn't, with nothing gradual in between. Risk_Category is functioning as a same-period outcome flag, not a forward-looking underwriting signal. That's the whole argument behind Discovery #1.

■ 5.5 By credit score band — the alternative that's already there

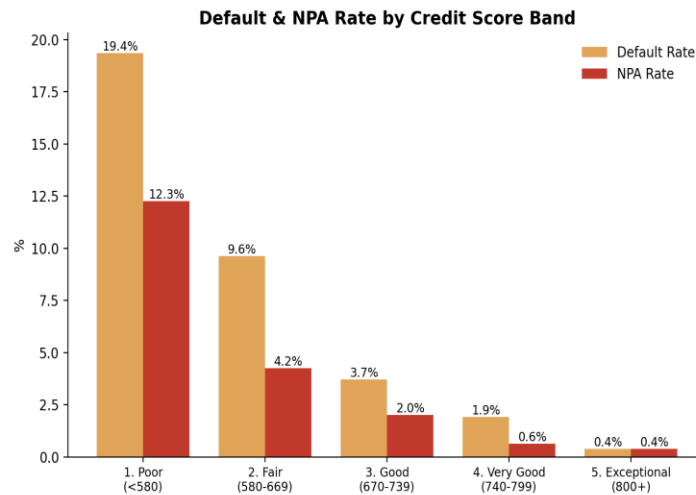


FIGURE 5.5 Default & NPA rate by standard credit-score band.

TABLE 5.4 — DEFAULT & NPA RATE BY CREDIT SCORE BAND

Credit Score Band	N	Default Rate	NPA Rate	Combined
1. Poor (<580)	155	19.35%	12.26%	31.61%
2. Fair (580–669)	1,059	9.63%	4.25%	13.88%
3. Good (670–739)	1,398	3.72%	2.00%	5.72%
4. Very Good (740–799)	945	1.90%	0.63%	2.53%
5. Exceptional (800+)	511	0.39%	0.39%	0.78%

This is the finding that makes Discovery #1 more than just a complaint. Where the mean-versus-mean comparison earlier showed overlapping, noisy distributions, **binning credit score into standard bureau bands produces a clean, monotonic gradient**: 31.6% combined failure in the bottom band down to 0.78% in the top one, a 40x spread, with every step down roughly halving the failure rate or better. This is exactly the forward-looking signal Risk_Category should have been. It costs nothing to build, since Credit_Score is already captured before a loan is ever approved.

■ 5.6 Does it change over time?

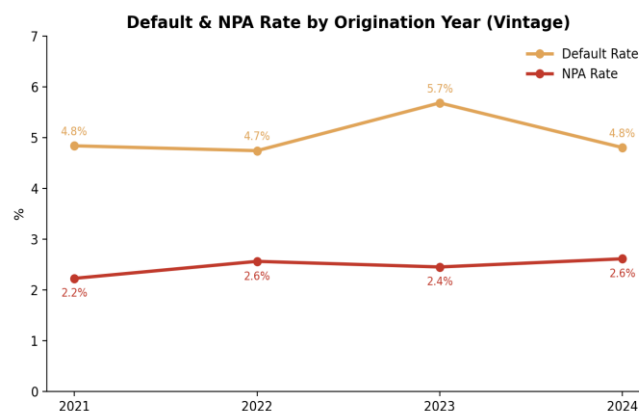


FIGURE 5.6 Default & NPA rate by origination year.

Not much. By year, Default and NPA rates are essentially flat. 2023 runs a little hot on Default (5.68% against 4.7–4.8% in the other three years), but there's no sustained drift up or down. Zoom into individual months and the rate gets noisy fast, swinging ± 5 to 7 points, which is exactly what you'd expect from only 70–100 loans originating in any given month, too small a sample to read much into. The yearly view is the one to trust, and it says underwriting quality has been stable across 2021–2024, not deteriorating.

■ 5.7 The 106 loans worth watching right now

106

Active or Closed loans still tagged "Critical" risk

The portfolio's most likely next defaults

76 Active and 30 Closed loans currently carry a Critical risk tag but haven't defaulted or gone NPA — yet. Since every single failure in the book so far has come out of exactly this bucket (5.4), these 106 loans are, statistically, the population most likely to produce the portfolio's next wave of Defaults and NPAs. They're worth a proactive look, not a wait-and-see one.

■ 5.8 Does the branch performance tier actually mean anything? (Q73)

TABLE 5.5 — APPROVAL, DEFAULT & NPA RATE BY PERFORMANCE_TIER

Performance_Tier	Branches	Approval Rate	Default Rate	NPA Rate
High	17	59.55%	4.46%	2.66%
Medium	28	58.22%	5.70%	2.44%
Low	21	55.12%	4.72%	1.98%

“High” tier branches approve a bit more freely than Low tier (59.55% vs. 55.12%), with no matching gain in Default rate (4.46% vs. 4.72%, within noise) and a higher NPA rate than Low. Medium tier, not Low, actually carries the worst combined outcome at 8.14%. Performance_Tier, in other words, is tracking throughput, not credit discipline or risk.

One more concentration worth flagging: MG Road Branch alone holds 7% of the bank's entire NPA count, with Bandra and Sector 17 close behind at 5% each (Q72).

06 Underwriting & Approval Alignment

Two patterns show up when you look at how approval and pricing decisions are actually made, versus how the risk data says they should be made.

■ 6.1 Pattern A — gatekeeping that doesn't flex

Sub-Prime approval sits at 27.8% company-wide, and **it barely moves by branch tier**: 27.3% at High-tier branches, 29.0% at Medium, 26.1% at Low. That kind of uniformity doesn't happen by individual judgment; it happens when there's a hard cut-off behind the scenes. And the segment still comes out the other side defaulting at 19.2% combined. The policy is doing a good job of controlling volume. It isn't doing much to control risk.

■ 6.2 Pattern B — pricing that hasn't caught up to risk

Interest rate does rise with risk category (9.36% average for Low, climbing to 10.98% for Critical), but that's a 162-basis-point spread sitting on top of a risk spectrum where Critical alone carries a 74.15% failure rate and Low, Medium, and High carry zero.

TABLE 6.1 — INTEREST RATE & CREDIT SCORE BY RISK CATEGORY

Risk Category	Avg. Interest Rate	Avg. Credit Score
Low	9.36%	800.7
Medium	9.81%	730.8
High	10.28%	677.2
Critical	10.98%	617.3

■ 6.3 What repayment behavior actually says

If you want to know where segment really shows up in day-to-day behavior, it's not in who pays late. Late-payment rates are nearly identical across segments: Sub-Prime 19.1%, Regular 21.1%, Mass Affluent 21.8%, Premium 22.5%. Ordinary arrears behavior isn't segment-driven. The effect only shows up at the point of full default, which matters for how collections should be run: early-stage tactics can be one-size-fits-all, but late-stage recovery strategy should not be.

TABLE 6.2 — PAYMENT STATUS DISTRIBUTION

Payment Status	Count	% of Payments
On-Time	9,100	70.0%
Late	2,741	21.1%
Partial	646	5.0%
Missed	513	3.9%

07 Payment & Collections Behavior

Everything so far has been about who gets a loan and how likely it is to go bad. This section is about what happens after: how payments actually flow in, and where they stop.

7.1 Where recoverability falls off a cliff (Q52)

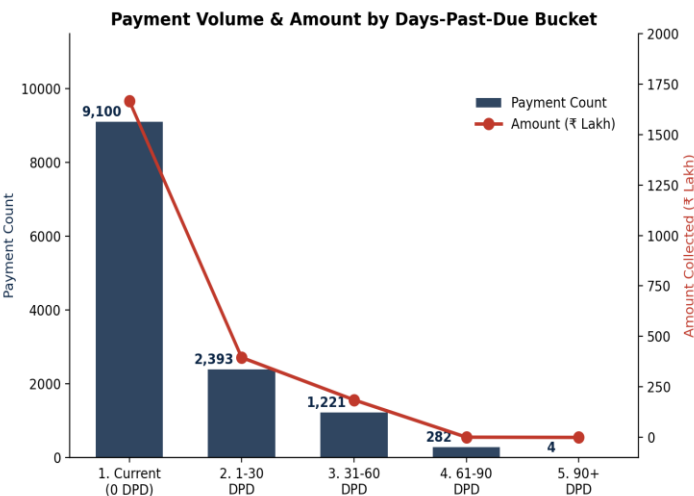


FIGURE 7.1 Payment volume and amount collected by days-past-due bucket.

Both payment volume and the amount actually collected drop sharply once a payment passes 30 days late. Of 13,000 payments, 9,100 (70.0%) are Current and another 2,393 (18.4%) are 1–30 DPD; together, that’s the vast majority of activity. By 61–90 DPD, only 282 payments remain, and they collect a combined ₹106,920, a rounding error next to the ₹166.7 lakh collected while Current. Collection’s effort today is clearly front-loaded, and the data backs that instinct: the 1–30 DPD window is where intervention still has something to work with.

7.2 Which products actually collect what they're owed (Q44)

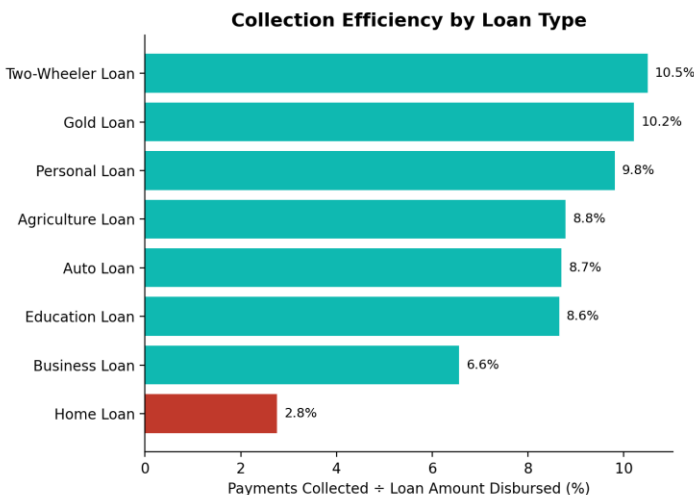


FIGURE 7.2 Payments collected as a share of loan amount disbursed, by loan type.

Two-Wheeler, Gold, Personal, Agriculture, Auto, and Education loans all sit in a reasonably tight 8.6–10.5% band. Business Loans dip to 6.56%. And **Home Loans fall to 2.76%**, by far the lowest of any product. That's not automatically alarming: Home Loans are also the largest-ticket, longest-tenure product in the book, and they take twice as long to approve (Section 4.5), so collecting only a small share of principal early in a long amortization schedule is somewhat expected. Treat this figure as a baseline rather than a red flag for now. It should be re-cut by loan age once Section 2.4's reconciliation issue is resolved, since right now there's no clean way to separate “expected for its stage” from “actually underperforming.”

08 Pricing Equity & Risk-Adjusted Return

A last set of checks on whether pricing and loan sizing track risk and need fairly, or whether they're leaving money or questions on the table.

- **Pricing hasn't caught up with risk:** that 162-basis-point spread between Low-risk (9.36%) and Critical-risk (10.98%) pricing is nowhere near proportional to the 32x gap in combined default rates between the best and worst customer segments (Premium 0.6% vs. Sub-Prime 19.2%). The riskiest loans in the book aren't paying for their own risk. The rest of the portfolio's margin is quietly absorbing the difference.
- **No sign of gender disparity, and that's worth saying plainly:** combined Default+NPA rates come out to 7.45% for male borrowers and 7.50% for female borrowers on the approved book, both essentially sitting on the 7.47% portfolio average. Worth stating clearly now and checking again as the book grows.
- **An unexpected gap by education level:** Doctorate holders receive the smallest average approved loan of any education tier, ₹7.75 lakh, roughly 28% below Graduates (₹10.72L), Post Graduates (₹10.65L), and High School-only borrowers (₹10.23L), who all cluster tightly together. Doctorate holders aren't the lowest-income group in the customer base, so this looks more like a loan-type or age effect than an income effect, worth a branch-level look to rule out an unstated cap.
- **The collections channel mix has room to improve:** UPI leads at 35.0%, then NEFT 19.5%, Auto-Debit 18.1%, Cheque 12.2%, Cash 9.9%, RTGS 5.3%. Auto-Debit's modest 18% share stands out because it's typically the channel most associated with reliable, on-time repayment. Under-enrollment here is a lever with a genuinely simple fix (Recommendation 4).

09 Strategic Recommendations

Five moves, ordered by how foundational they are. The first two aren't optional extras: nearly every other number in this report either feeds from Risk_Category or from the Payments/Loans reconciliation, so fixing those two first is what makes everything after them trustworthy.

RECOMMENDATION 1

Rebuild Risk_Category around the credit-band gradient already in the data

WHAT WE FOUND Risk_Category shows 0% Default and NPA in Low, Medium, and High combined, and 74.15% combined failure in Critical alone (5.4). It separates loans that have already failed from those that haven't, which isn't underwriting. Meanwhile, standard credit-score bands (5.5) already show a clean, monotonic 40x gradient using nothing but data available at the moment of application.

WHAT TO DO Build a scorecard model starting from the credit-band gradient as a baseline feature, layering in income, occupation, loan type, and tenure, validated out-of-time. Call it what it is: a “Pre-Approval Risk Score,” distinct from any post-funding performance tag, and retire Risk_Category as an underwriting input until this is live.

WHY IT MATTERS A genuinely predictive score lets the bank price and gate risk before capital goes out the door, not after. And the credit-band evidence means the bank isn't starting from zero. The raw material for a 40x risk gradient is already sitting in the data.

RECOMMENDATION 2

Treat the Payments/Loans mismatch as a data-integrity incident, not a reconciliation task

WHAT WE FOUND Q13's per-loan check shows all 4,068 approved loans fail reconciliation between Amount_Repaid and actual summed Payments: not a gap, a universal failure. Amount_Repaid averages 5.25x the evidenced payment total, correlating at just 0.75, and 499 approved loans (12.3%) have no payment history at all, including 130 of the 304 Default+NPA loans.

WHAT TO DO Escalate to Finance and Data Engineering as a data-integrity incident. Trace the actual source system populating Amount_Repaid, resolve the orphan Loan_ID (LN0005644) referenced by two stray Payments rows, and freeze use of Amount_Repaid, Outstanding_Balance, or any collections-efficiency figure derived from them in external or board reporting until this is resolved.

WHY IT MATTERS Restores confidence in every collections and outstanding-balance figure downstream: Sections 4, 5, and 7 of this report, and the ₹295.17 crore outstanding-balance figure specifically.

RECOMMENDATION 3

Replace the flat Sub-Prime cut-off with underwriting that actually varies

WHAT WE FOUND Sub-Prime approval sits at a nearly identical 26–29% across every branch tier (High, Medium, and Low alike), which points to a blanket policy rule rather than case-by-case judgment. The segment still defaults at 12.88% (plus 6.33% NPA), the highest combined rate anywhere in the book.

WHAT TO DO Add a second underwriting layer specifically for Sub-Prime applicants: deeper income verification, smaller initial loan sizing, mandatory Auto-Debit enrollment as a condition of approval, instead of one uniform accept/reject line.

WHY IT MATTERS Targets the segment responsible for the largest share of the 304 Default+NPA loans, without simply rejecting more of the 80.8% of approved Sub-Prime loans that are performing just fine.

RECOMMENDATION 4

Price for the risk that's actually there, and get more of the book onto Auto-Debit

WHAT WE FOUND A 162-basis-point pricing spread is doing very little work against a 32x combined default-rate gap between the best and worst segments. Meanwhile Auto-Debit (the payment channel most associated with reliable repayment) covers only 18.1% of transactions.

WHAT TO DO Once Recommendation 1's rescored model is live, widen the risk-based pricing tiers to actually reflect the gap. In parallel, run a targeted Auto-Debit conversion push aimed first at the 106 Active/Closed loans still carrying a Critical tag (5.7). Those are the loans statistically closest to the portfolio's next default.

WHY IT MATTERS Better risk-adjusted margin on the highest-risk performing loans, and a direct de-risking of the flight-risk cohort before it converts to Default or NPA.

RECOMMENDATION 5

Move collections earlier, using the DPD cliff as the trigger

WHAT WE FOUND Recoverable amounts fall off sharply after 30 days past due (7.1): by 61–90 DPD, only ₹106,920 is coming back across 282 payments, against ₹166.7 lakh collected while Current. No process currently in place is preventing that cliff.

WHAT TO DO Build an automated 1–30 DPD outreach trigger (the window where recovery is still realistic) rather than waiting for later-stage collections to kick in. Separately, once Recommendation 2 is resolved, re-run the Home Loan 2.76% collection-efficiency figure (7.2) segmented by loan age, to confirm it reflects tenure rather than a genuine shortfall on the bank's largest asset class.

WHY IT MATTERS Puts limited collections resources where they can still change the outcome, and closes the loop on whether the Home Loan book (51% of total approved value) needs its own collections strategy or is simply performing as expected for where it sits in its lifecycle.

metrics are scoped to dbo.Approved_Loans. Figures reflect the full dataset as extracted; the reconciliation gap in Recommendation 2 should be resolved before Amount_Repaid-derived figures are used for external or board-level reporting.